OPERATING SYSTEMS

MCA — PYQ-Style Practice Question Paper

Time: 3 Hours Maximum Marks: 76

Instructions: Attempt all questions. Draw neat diagrams and show complete steps for numerical problems.

Q1. [5 marks] Define an Operating System. Explain any five major functions or services provided by an OS.

Q2. [5 marks] Explain the different process states with a neat process-state transition diagram. Also explain the role of the Process Control Block (PCB).

Q3. [5 marks] What is a system call? Explain different categories of system calls with suitable examples.

Q4. [5 marks] Explain FCFS, SJF, and Round Robin CPU scheduling. Compare their advantages and disadvantages.

Q5. [6 marks] Consider the following processes. Calculate Waiting Time and Turnaround Time using SJF (non-preemptive) and draw the Gantt chart.

Process	Arrival Time	Burst Time
P1	0	8
P2	1	4
P3	2	2
P4	3	1

Q6. [5 marks] What is process synchronization? Explain critical section, race condition, mutex, and semaphore.

Q7. [5 marks] Define deadlock. Explain the four necessary conditions for deadlock and discuss any two methods for handling deadlocks.

Q8. [5 marks] Explain the Banker’s Algorithm for deadlock avoidance. What is meant by a safe state and safe sequence?

Q9. [5 marks] Explain paging with a suitable example. Differentiate between logical address and physical address.

Q10. [6 marks] For the page reference string 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5, assuming 3 page frames, calculate the number of page faults using FIFO and LRU.

Q11. [5 marks] What is virtual memory? Explain demand paging, page fault, and thrashing.

Q12. [5 marks] Explain the three major file allocation methods: contiguous allocation, linked allocation, and indexed allocation. Compare their advantages and disadvantages.

Q13. [5 marks] Explain disk scheduling. Compare FCFS, SSTF, SCAN, and C-SCAN algorithms.

Q14. [5 marks] A disk has cylinders numbered 0–199. The disk head is currently at 50. The request queue is 82, 170, 43, 140, 24, 16, 190. Calculate the total head movement using SSTF.

Q15. [4 marks] Explain: (a) Authentication vs Authorization (b) Protection in Operating Systems (c) Virtualization and Virtual Machines (d) Basic Linux/Unix OS concepts.